
Calculus I

Explanations, Problems & Solutions (Solutions)

Written by :

Sihoo Lee % Hojun Lee



KOREA SCIENCE ACADEMY OF KAIST

2026-04-12

Preface

1 Teacher

Hyojin Kim 김효진 선생님

hyojinkim@ksa.kaist.ac.kr

- no math questions here
- otherwise all accepted!

1.1 Office Hours

- Tell before visiting.

2 Syllabus

MainTextbook: Stewart Calculus

- Thomas Calculus (supplements)
 - 김홍중 Calculus (SNU textbook)
 - KAIST Calculus
-

Sihoo Lee % Hojun Lee
Korea Science Academy of KAIST

Contents

Chapter 00

Functions and Limits

This chapter formalizes the concept of the limit, moving from intuitive approximations to rigorous mathematical definitions including the precise Epsilon-Delta definition. It provides the algebraic rules (Limit Laws) necessary to calculate limits without graphing, methods for handling indeterminate forms, and defines continuity to describe functions without breaks or holes. These concepts culminate in the Intermediate Value Theorem, a powerful tool used to prove the existence of roots and solutions within specific intervals, setting the stage for the definition of the derivative.

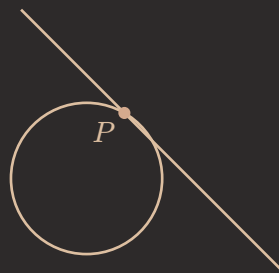
The Tangent & Speed Problem

1 The tangent Problem

Definition | Tangent Line

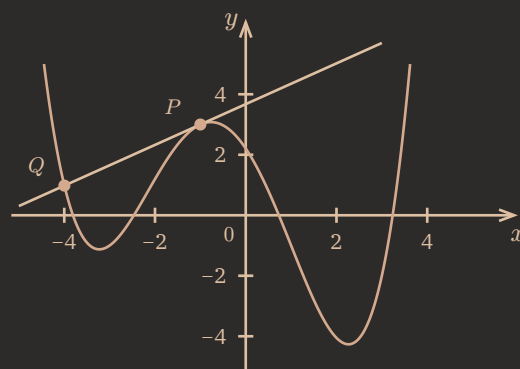
a **tangent line** to a curve at a point is the line that just touches the curve at that point and best approximates the curve near that point. A tangent line need not intersect at only one point; it may cross the curve or meet it again elsewhere

Example | Circle vs Function



On circles, the tangent line can be easily defined following *Euclid*. The tangent line here at point P is a line that only meets the circle **once** on P .

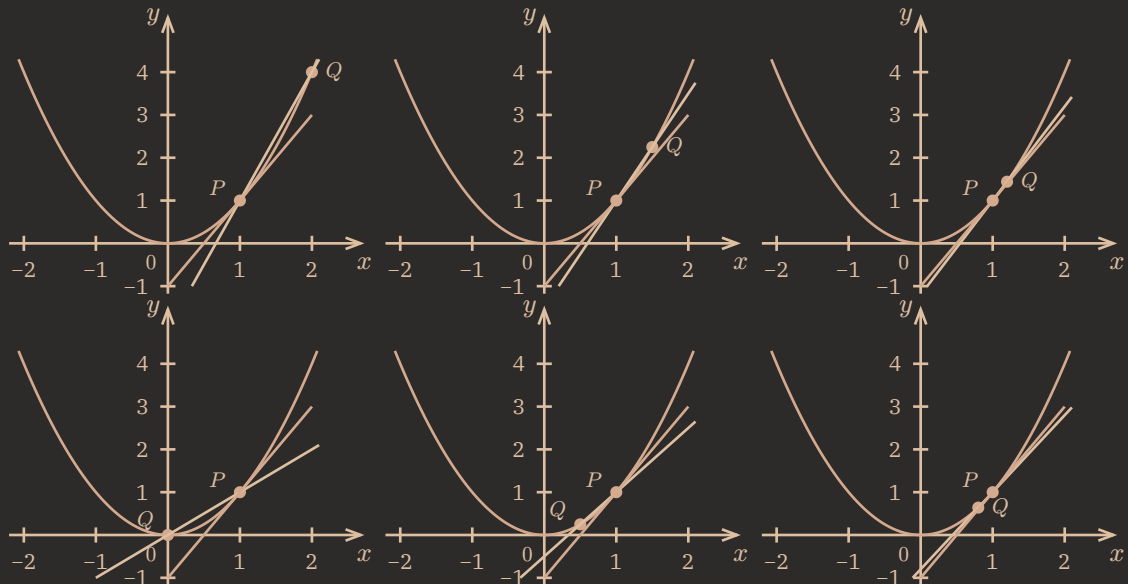
However, this convenient definition cannot be generalized to all functions.



As you can see, The tangent line on P also meets with the function at Q too. We cannot simply use Euclidean definition here. How should we define this?

Example |

Now we try to find an equation of the tangent line to the parabola $y = x^2$ at the point $P(1, 1)$.



It seems like as P approaches the approximation gets better.

The slope is

$$m_{PQ} = \frac{x^2 - 1}{x - 1}$$

Thus,

$$\lim_{P \rightarrow Q} \frac{x^2 - 1}{x - 1} = (\text{tangent slope})$$

How do we calculate? NGD.

P_x	2	1.5	1.1	1.01	1.001
m_{PQ}	3	2.5	2.1	2.01	2.001

P_x	0.999	0.99	0.9	0.5	0
m_{PQ}	1.999	1.99	1.9	1.5	1

Looks like 2.

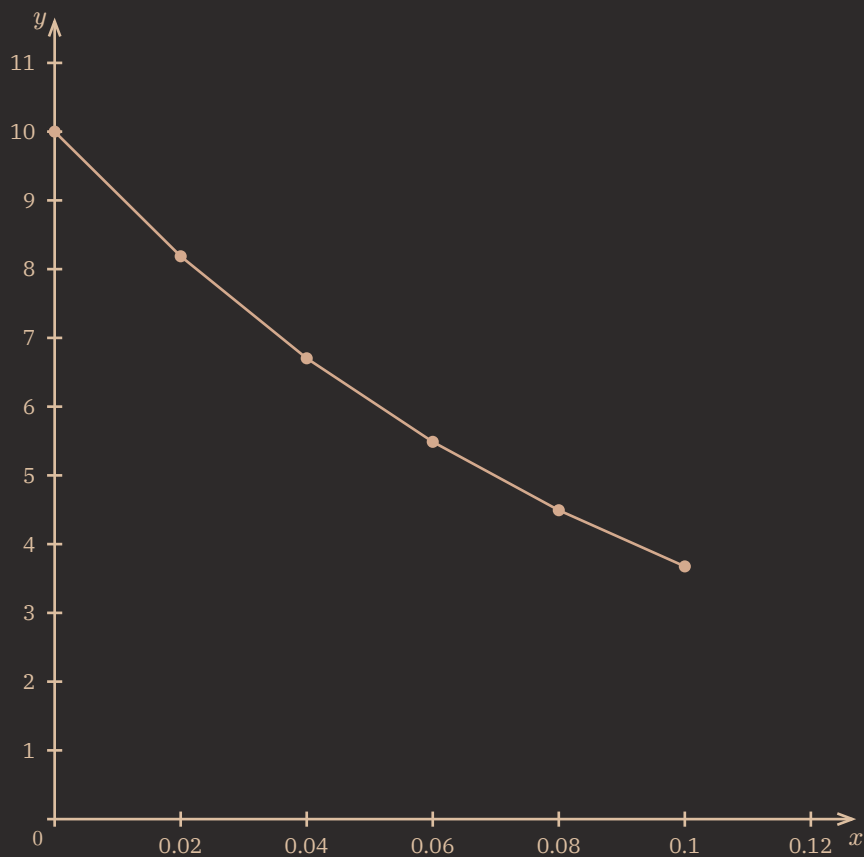
$$\therefore y = 2x - 1$$

Example | pulse laser

A pulse laser operates by storing charge on a capacitor and releasing it suddenly when the laser is fired. The data in the table describe the charge Q remaining on the capacitor (measured in coulombs) at time t (measured in seconds after the laser is fired). Use the data to draw the graph of this function and estimate the slope of the tangent line at the point where $t = 0.04$.

(Note: The slope of the tangent line represents the electric current flowing from the capacitor to the laser [measured in amperes].)

t	0	0.02	0.04	0.06	0.08	0.1
Q	10	8.187	6.703	5.488	4.493	3.676



Average adjacent slopes. (Justification: Taylor and... will someday discovered)

Chapter 00.05

The Limit of a Function

In this section we introduce intuitive definitions of limits. Further detail on Ch 1.7

Definition |

Suppose $f(x)$ is defined for x *near* the number a , meaning that f is defined on some open interval including a .

Then we write

$$\lim_{\{x \rightarrow a\}} f(x) = L$$

We read this as “the limit of $f(x)$ as x approaches a is L ”. This claim must be preceded by the statement that as $f(x)$ gets arbitrarily closer and closer to L as x gets closer and closer to a .

Note |

We don't care about $f(a)$. Even if it do not exist or rebel!

Example | Catastrophic cancellation / Round-off error

Let's evaluate the values of

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$$

t	± 1	± 0.5	± 0.1	± 0.05	± 0.01
$\frac{\sqrt{t^2 + 9} - 3}{t^2}$	0.1622777	0.1655251	0.1666204	0.1666551	0.1666662

We can see the results are approaching to 0.16666...

So it looks like the value of the limit is $\frac{1}{6}$.

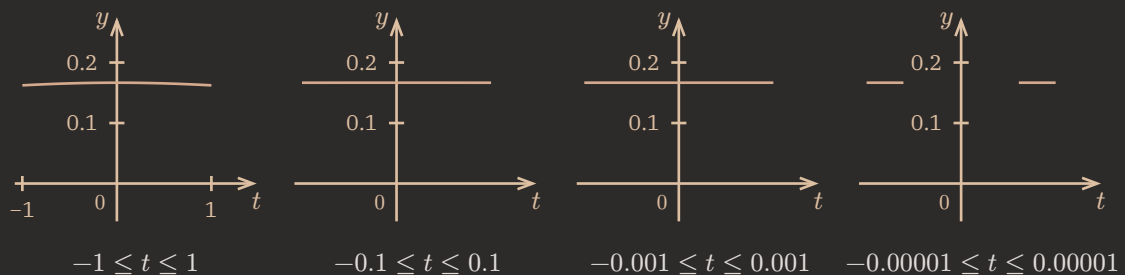
To make it sure, Let's try plugging in even smaller number.

t	± 0.001	± 0.0001	± 0.00001	± 0.00000001
-----	-------------	--------------	---------------	------------------

$\frac{\sqrt{t^2 + 9} - 3}{t^2}$	0.166666662	0.166666668	0.166666668	0
----------------------------------	-------------	-------------	-------------	---

?????

Is the real limit 0? Actually, the reason is imperfect algorithm of calculator. Since calculator is sort of a computer and it only can compute discrete values, very briefly speaking, it somehow works like *flooring* the values of infinite (or very many) numbers in decimal space. This causes the calculator think $\sqrt{t^2 + 9} - 3 = 0$ when t is sufficiently small.



You can also see corrupted part on graphing calculator in extremely small values.

Example | Geuss the value of limits

Evaluate the following: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$

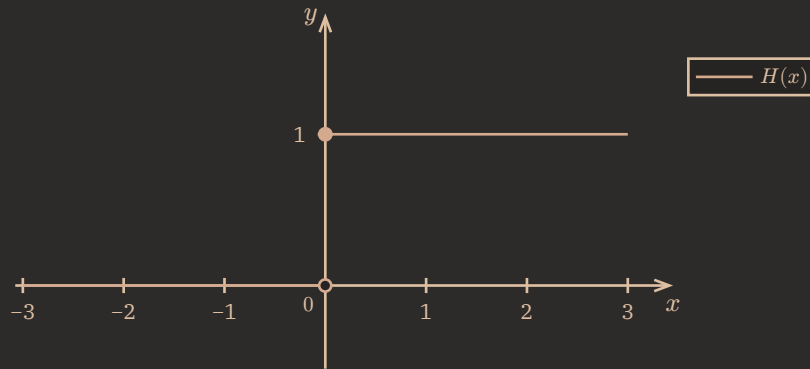
Remember: $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$, $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$

Example |

Heaviside Function:

$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

What is $\lim_{x \rightarrow 0} H(x)$?



Definition | Intuitive Definition of One-Sided Limits

Suppose $f(x)$ is defined whilst x is near a .
Then we write

$$\lim_{\{x \rightarrow a^+\}} f(x) = L_1$$

and

$$\lim_{\{x \rightarrow a^-\}} f(x) = L_2$$

We read these as “the limit of $f(x)$ as x approaches a from the right is L_1 ”
and “the limit of $f(x)$ as x approaches a from the left is L_2 ”.

This claim must be preceded by the statement that as $f(x)$ gets arbitrarily closer and closer to L_1 as x gets closer and closer to a from the right, and similarly for L_2 from the left.

Here, we call L_1 as the **right-hand limit** and L_2 as the **left-hand limit**.

The value of limits only exist when both left- and right-hand limits are equal.

Thus the limit here does not exist.

- With the basic idea, we can now consider the bigger problem : can limits **fail** to exist?

Example | Uh Oh

Investigate the following :

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right)$$

Solution 1 |

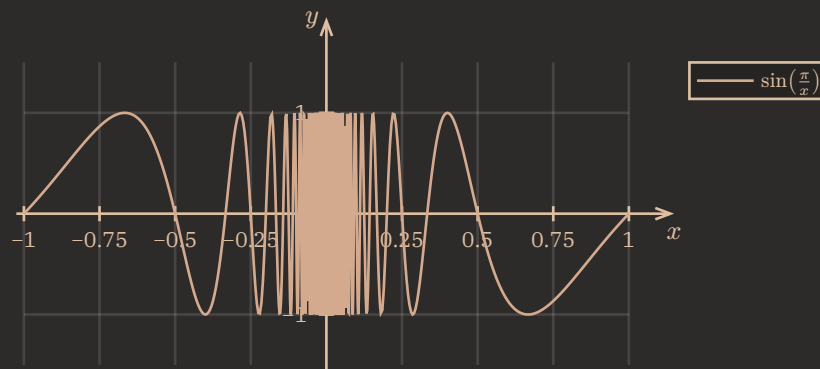
To sketch the graph of $f(x) = \sin\left(\frac{\pi}{x}\right)$, note first that f is an odd function, thus its graph is symmetric with respect to the origin. As reaching 0, the oscillation gets faster.

x	$\sin\left(\frac{\pi}{x}\right)$
1	0
0.1	0.58778525
0.01	0.95105652
0.001	0.30901699
0.0001	0.98768834

By evaluating $\sin\left(\frac{\pi}{x}\right)$ at values of x that get closer and closer to 0, we can see that the function does not approach a single value. Instead, it oscillates between -1 and 1 . Therefore, we can conclude that:

$$\lim_{x \rightarrow 0} \sin\left(\frac{\pi}{x}\right)$$

does not exist.



Example | Uh Oh 2

Find

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} \right)$$

if it exists

Solution 1 |

x	1	0.1	0.01	0.001	0.0001
$\frac{1}{x^2}$	1	100	10000	1000000	100000000

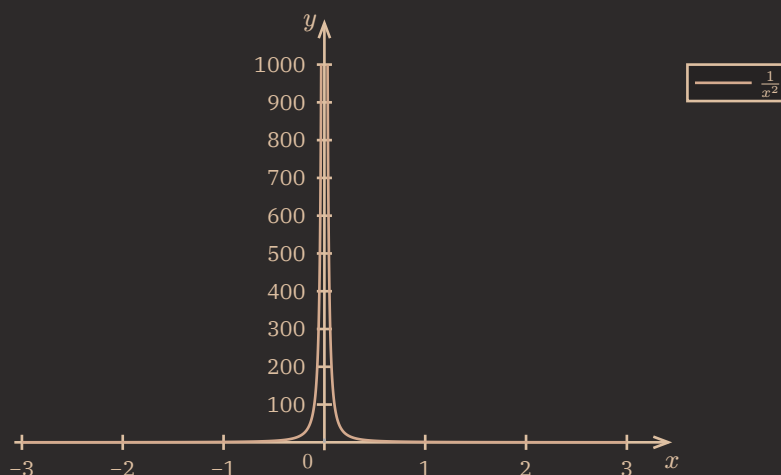
By evaluating $\frac{1}{x^2}$ at values of x that get closer and closer to 0, we can see that the function grows without bound. Therefore, we can conclude that:

Notation | Infinity

When the value increase boundlessly,

$$\lim_{x \rightarrow 0} \left(\frac{1}{x^2} \right) = \infty$$

(which mathematically means that the limit does not exist in the real number system)



Another example is the Dirichlet function, which will be introduced in Section 8. Of course, one-side limit that do not exist also exists.

Definition |

The vertical line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true.

$$\begin{array}{ll} \lim_{x \rightarrow a^-} f(x) = \infty & \lim_{x \rightarrow a^-} f(x) = -\infty \\ \lim_{x \rightarrow a^+} f(x) = \infty & \lim_{x \rightarrow a^+} f(x) = -\infty \end{array}$$

Thus if only one side is approaching infinity, it is considered as a vertical asymptote.

Chapter 00.06

Calculating Limits using the Limit Laws

Theorem | The Laws of Limits

Suppose that c is a constant and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist. Then, the following properties hold:

1. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$ Sum Law
2. $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$ Difference Law
3. $\lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot \lim_{x \rightarrow a} f(x)$ Constant Multiple Law
4. $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$ Product Law
5. $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$, provided that $\lim_{x \rightarrow a} g(x) \neq 0$ Quotient Law

Theorem |

$$\lim_{x \rightarrow a} c = c$$

$$\lim_{x \rightarrow a} x = a$$

We just take it and accept it.

Constant Multiple Law should be derived from here fundamentally.

Example |

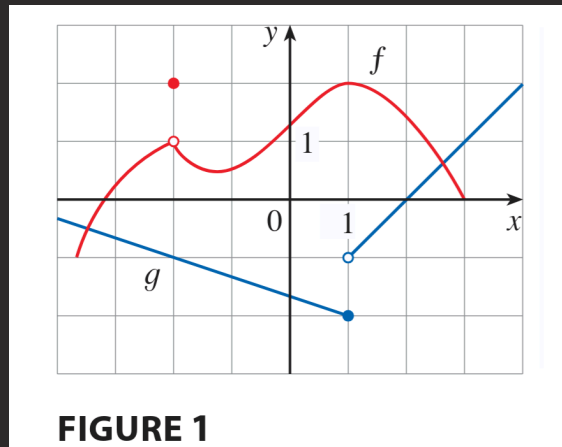


FIGURE 1

$$\lim_{x \rightarrow -2} f(x) + g(x) \quad \lim_{x \rightarrow 1} f(x)g(x) \quad \lim_{x \rightarrow 2} \frac{f(x)}{g(x)}$$

Solution 1 |

Note |

Even when $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ do not exist, $\lim_{x \rightarrow a} f(x) + g(x)$ may exist.

Proof |

$$f(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

$$g(t) = \begin{cases} 1 & \text{if } t < 0 \\ 0 & \text{if } t \geq 0 \end{cases}$$

Try finding more complex antiexample! (Exercise)

Theorem | The Power Law of Limits

Suppose that $\lim_{x \rightarrow a} f(x) = L$ and n is a positive integer. Then, $\lim_{x \rightarrow a} [f(x)]^n = \left[\lim_{x \rightarrow a} f(x) \right]^n = L^n$

Theorem | The Root Law of Limits

Suppose that $\lim_{x \rightarrow a} f(x) = L$ and n is a positive integer. Then, $\lim_{x \rightarrow a} [f(x)]^{\frac{1}{n}} = \left[\lim_{x \rightarrow a} f(x) \right]^{\frac{1}{n}} = L^{\frac{1}{n}}$, provided that if n is even, then $L \geq 0$

Example |

$$(a) \lim_{x \rightarrow -2} (2x^2 - 3x + 4) \quad (b) \lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{2x^2 - 3x + 4}$$

Solution 1 |

$$\begin{aligned} \lim_{x \rightarrow -2} (2x^2 - 3x + 4) &= \lim_{x \rightarrow -2} 2x^2 - \lim_{x \rightarrow -2} 3x + \lim_{x \rightarrow -2} 4 \\ &= 2 \left(\lim_{x \rightarrow -2} x \right)^2 - 3 \lim_{x \rightarrow -2} x + 4 = 2 \cdot (-2)^2 - 3 \cdot (-2) + 4 = 18 \end{aligned}$$

Similar for the rest.

Theorem | Direct Substitution Property

If f is a polynomial or a rational function and a is in the domain, then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Proof on practice problems.

Example | Former NGD Slope

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1}$$

Solution 1 |

Direct substitution is not acceptable because the denominator becomes 0. Since x is not technically 1,

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x + 1)(\cancel{x - 1})}{\cancel{x - 1}} = \lim_{x \rightarrow 1} x + 1 = 2$$

Note |

Technically $\frac{x^2-1}{x-1}$ is not same as $x + 1$ because of hole

Example | Conjugates

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2}$$

Solution 1 |

Multiply conjugate then root rule + direct substitution rule.

Note | Warning

Do not take the limit of only one part.

$$\lim_{t \rightarrow 0} \frac{\sqrt{t^2 + 9} - 3}{t^2} = \lim_{t \rightarrow 0} \frac{\lim_{t \rightarrow 0} (\sqrt{t^2 + 9} - 3)}{t^2} = \lim_{t \rightarrow 0} \frac{0}{t^2} = 0$$

bruh

Example | Not by Contradiction

$$\lim_{x \rightarrow 1} \frac{f(x) - 8}{x - 1} = 10$$

Find $\lim_{x \rightarrow 1} f(x)$

Solution 1 |

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} \left[\frac{f(x) - 8}{x - 1} \right] (x - 1) + 8 = 8$$

Example | Gauss Function

For each real number x there exists a unique integer n and a number α with $0 \leq \alpha < 1$ such that

$$x = n + \alpha$$

The **Greatest integer function (Floor function)** is defined by

$$[[x]] = n$$

Show that $\lim_{x \rightarrow 3}$ does not exist

Solution 1 |

Theorem | The Squeeze Theorem

Theorem |

If $f(x) \leq g(x)$ when x is near a (not necessary on a) and the limits exist, then

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$$

This stands on one-sided limits. Not proven here.

If $f(x) \leq g(x) \leq h(x)$ when x is near a (except possibly at a) and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L \implies \lim_{x \rightarrow a} g(x) = L$$

Example | The Squeeze Theorem..?

If $f(x) < g(x)$ when x is near a (not necessary on a) and the limits exist, then

$$\lim_{x \rightarrow a} f(x) < \lim_{x \rightarrow a} g(x)$$

Will this be true?

Solution 1 |

$$f(x) = 0 \quad g(x) = x^2$$

Duh. always put equal for limits.

Example | Squeeze Theorem Example

Show that $\lim_{x \rightarrow 0} x^3 \sin\left(\frac{1}{x}\right) = 0$

Solution 1 |

$$\left| x^3 \sin\left(\frac{1}{x}\right) \right| = |x|^3 \left| \sin\left(\frac{1}{x}\right) \right| \leq |x|^3, \quad x \neq 0$$

Chapter 00.07

The Precise Definition of a Limit

1 Historical Motivation

In the previous section,

$$\lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x+1)\cancel{(x-1)}}{\cancel{x-1}} = \lim_{x \rightarrow 1} (x+1) = 2$$

We informally say x *approaches* 1. What is this exactly?

Philosopher George Berkeley:

“Havent’t you said $x - 1$ is 0? Why is it sometimes treated as 0 and sometimes not?” (The ghost number problem)

In 19th century, Cauchy introduced limits as the central concept of calculus, and later Weierstrass formulated modern $\varepsilon - \delta$ definition.

Then we say that $f(x)$ approaches L as x approaches a , we think of **error** decreasing, i.e., $|f(x) - L|$ decreases.

2 The Precise Definition

Definition | ε - δ

Let f be a function defined on some open interval that contains the number a , except possibly at a itself. Then we say that x approaches a

$$\forall \varepsilon > 0 [\exists \delta > 0 [0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon]]$$

Intuitively, we can think of sort of a *aggressive* algorithm,

1. Critic suggests an acceptable error ε
2. Actor provides an appropriate range δ to suffice the suggested error.
3.
4. If we can prove that suggesting appropriate range is possible for any given error, we accept the limit value.

Definition | ε - M

$$\forall M > 0 [\exists \delta > 0 [0 < |x - a| < \delta \implies f(x) > M]]$$

3 Limit Laws Revisited

Theorem | The Laws of Limits

Suppose that c is a constant and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist. Then, the following properties hold:

1. $\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$ Sum Law
2. $\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$ Difference Law
3. $\lim_{x \rightarrow a} [c \cdot f(x)] = c \cdot \lim_{x \rightarrow a} f(x)$ Constant Multiple Law
4. $\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$ Product Law
5. $\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}$, provided that $\lim_{x \rightarrow a} g(x) \neq 0$ Quotient Law

Proof |

$$\lim_{x \rightarrow a} f(x) = L, \quad \lim_{x \rightarrow a} g(x) = M$$

Then we know that there exists delta

$$0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$$

4 Negation of a limit

We know the rigorous definition of the limit of a function:

$$f : D \rightarrow R, \quad a \in D$$

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \varepsilon > 0 \exists \delta > 0 \ 0 < |x - a| < \delta \rightarrow |f(x) - L| < \varepsilon$$

Thus the negation is

$$\begin{aligned} & \neg \left[\lim_{x \rightarrow a} f(x) = L \iff \forall \varepsilon > 0 \exists \delta > 0 \ 0 < |x - a| < \delta \rightarrow |f(x) - L| < \varepsilon \right] \\ & \iff \exists \varepsilon > 0, \forall \delta > 0, \exists x (0 < |x - a| < \delta \wedge |f(x) - L| \geq \varepsilon) \end{aligned}$$

Example |

Using epsilon delta argument, show that $\lim_{x \rightarrow 2} x^3 \neq 1$

Solution 1 |

Let $\varepsilon = 1$, For any $\delta > 0$, choose $n \in \mathbb{N}$ such that $\frac{1}{n} < \delta$, and let $x = 2 + \frac{1}{n}$, Then $0 < |x - 2| < \delta$, and

$$|x^3 - 1| =$$

Example | Dirichlet Function

If the function f is defined by

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q} \\ 1 & \text{if } x \in \mathbb{R} \setminus \mathbb{Q} \end{cases}$$

Show that $\lim_{x \rightarrow 0} f(x)$ does not exist.

Solution 1 | Focusing on negation

Suppose that it exists.

First we show that $L = 0$. Let $\varepsilon = \frac{|L|}{2}$ For any δ , we can choose $x \neq 0$ such that it is 0. (Density argument)

Same applies to $L = 1$. The limit value is unique. contradiction.

Definition | Sequential Characterization

A sequence $\{a_n\}$ has the limit L and we write

$$\lim_{n \rightarrow \infty} a_n = L \quad n \rightarrow \infty \implies a_n \rightarrow L$$

We have similar epsilon delta like statement here too. Also, we can think of it as

Todo: fill it based on the ppt.

Chapter 00.08

Continuity

1 Continuity

Intuitively, continuity can be simply defined as “a function that is connected”...but we don't deal with intuition here. We need a more “precise” definition of continuity.

Definition | Continuity at a Point

A function f is said to be continuous at a point $x = a$ if the following three conditions are satisfied :

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a} f(x)$ exists.
3. $\lim_{x \rightarrow a} f(x) = f(a)$

Equivalently, for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that

$$|x - a| < \delta \implies |f(x) - f(a)| < \varepsilon$$

If these conditions are not met, we call the function is “discontinuous” at $x = a$, or has a “discontinuity” at $x = a$.

Example | Continuity Example

Identify if the functions are discontinuous.

(a) $f(x) = \frac{x^2 - x - 2}{x - 2}$

(b) $f(x) = \begin{cases} \frac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$

Solution (a) |

The function is discontinuous at $x = 2$. This is because $f(2)$ is not defined, since the denominator becomes 0 at this point. Therefore, the first condition for continuity is not satisfied.

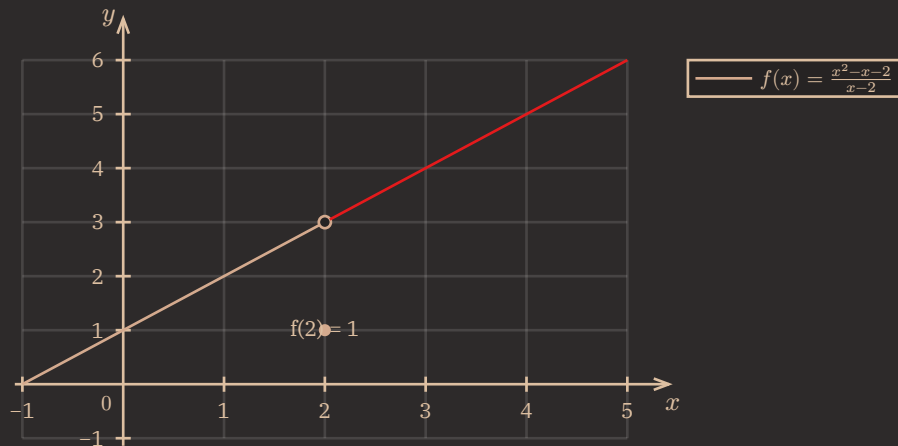
Solution (b) |

The function is continuous at $x = 2$. Let's check the three conditions:

1. $f(2) = 1$, so the first condition is satisfied.

2. To find $\lim_{x \rightarrow 2} f(x)$, we simplify the expression for $x \neq 2$: $f(x) = \frac{x^2 - x - 2}{x - 2} = \frac{(x-2)(x+1)}{x-2} = x + 1$ for $x \neq 2$. Therefore, $\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x + 1) = 3$. So the second condition is satisfied.
3. However, $\lim_{x \rightarrow 2} f(x) = 3$ and $f(2) = 1$, so the third condition is not satisfied.

Thus, the function is discontinuous at $x = 2$.



As we did with limits, we can define the left and right continuities for a function too.

Definition | Left Continuity at a Point

A function f is said to be left continuous at a point $x = a$ if the following three conditions are satisfied :

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a^-} f(x)$ exists.
3. $\lim_{x \rightarrow a^-} f(x) = f(a)$

Definition | Right Continuity at a Point

A function f is said to be right continuous at a point $x = a$ if the following three conditions are satisfied :

1. $f(a)$ is defined.
2. $\lim_{x \rightarrow a^+} f(x)$ exists.
3. $\lim_{x \rightarrow a^+} f(x) = f(a)$

Right side limits appear frequently in probability theory, while left continuity are less commonly used.

Using this info, we can create the following definition :

Definition | Continuity on an Interval

A function f is said to be continuous on an interval I if it is continuous at every point in I .

- If I is a closed interval $[a, b]$, then we also need to check that f is left continuous at $x = b$ and right continuous at $x = a$.

Note | Remark

If f is defined only on one side of an endpoint of the interval, we understand continuous at the endpoint to mean continuous from the right or left continuity.

Example | Continuity on an Interval Example

Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous in the interval $[-1, 1]$.

Solution 1 |

If $-1 < a < 1$, then using the Limit Laws from Section 1.6, we have :

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} (1 - \sqrt{1 - x^2}) = 1 - \lim_{x \rightarrow a} \sqrt{1 - x^2} = 1 - \sqrt{1 - a^2} = f(a)$$

. Thus, f is continuous at every point in the open interval $(-1, 1)$. Now, we need to check the endpoints $x = -1$ and $x = 1$.

- At $x = -1$:

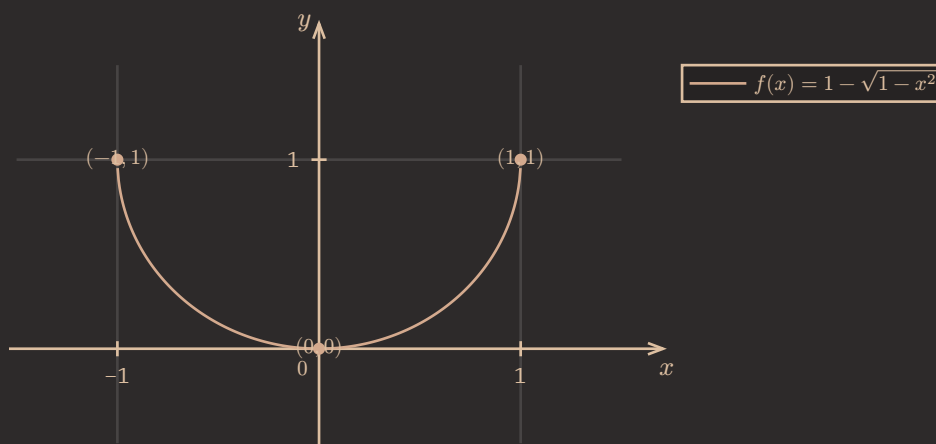
$$\begin{aligned} \lim_{x \rightarrow -1^+} f(x) &= \lim_{x \rightarrow -1^+} (1 - \sqrt{1 - x^2}) \\ &= 1 - \lim_{x \rightarrow -1^+} \sqrt{1 - x^2} \\ &= 1 - \sqrt{1 - (-1)^2} \\ &= 1 - 0 \\ &= 1 = f(-1) \end{aligned}$$

. Thus, f is right continuous at $x = -1$.

- At $x = 1$:

$$\begin{aligned}
 \lim_{x \rightarrow 1^-} f(x) &= \lim_{x \rightarrow 1^-} (1 - \sqrt{1 - x^2}) \\
 &= 1 - \lim_{x \rightarrow 1^-} \sqrt{1 - x^2} \\
 &= 1 - \sqrt{1 - 1^2} \\
 &= 1 - 0 \\
 &= 1 = f(1)
 \end{aligned}$$

. Thus, f is left continuous at $x = 1$. Therefore, f is continuous on the closed interval $[-1, 1]$.



2 Properties of Continous Functions

Just like limits, continuous functions have some properties that we can use to solve problems more easily.

Theorem | Properties of Continuous Functions

If f and g are continuous at $x = a$, then the following functions are also continuous at $x = a$:

1. $f + g$
2. $f - g$
3. cf , where c is any constant
4. fg
5. $\frac{f}{g}$, provided that $g(a) \neq 0$
6. $[f(x)]^n$, where n is a positive integer (derived from rule 4)

Proof | Rule 1

$$\lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) = f(a) + g(a) = (f + g)(a)$$

Thus, since $f(x)$ and $g(x)$ are continuous at $x = a$, $f + g$ is also continuous at $x = a$.

Proof | Rule 2

$$\lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x) = f(a) - g(a) = (f - g)(a)$$

Thus, since $f(x)$ and $g(x)$ are continuous at $x = a$, $f - g$ is also continuous at $x = a$.

Proof | Rule 3

$$\lim_{x \rightarrow a} [cf(x)] = c * \lim_{x \rightarrow a} f(x) = c * f(a) = (cf)(a)$$

Thus, since $f(x)$ is continuous at $x = a$, cf is also continuous at $x = a$.

Proof | Rule 4

$$\lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) * \lim_{x \rightarrow a} g(x) = f(a) * g(a) = (fg)(a)$$

Thus, since $f(x)$ and $g(x)$ are continuous at $x = a$, fg is also continuous at $x = a$.

Proof | Rule 5

$$\lim_{x \rightarrow a} \left[\frac{f(x)}{g(x)} \right] = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{f(a)}{g(a)} = \left(\frac{f}{g} \right)(a)$$

Thus, since $f(x)$ and $g(x)$ are continuous at $x = a$ and $g(a) \neq 0$, $\frac{f}{g}$ is also continuous at $x = a$.

Proof | Rule 6

This rule is derived from Rule 4. Since $f(x)$ is continuous at $x = a$, using Rule 4 repeatedly, we can show that $[f(x)]^n$ is also continuous at $x = a$.

Theorem |

- polynomials
- root functions
- rational functions
- trigonometric functions

these are all continuous. trig functions will be revisited.

You may have noticed that we didn't even care about continuity during polynomials. This is because of the following theorem :

Theorem | Continuity of Polynomials and Rational Functions

1. Every polynomial function is continuous for all real numbers; that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$
2. Every rational function is continuous at every point in its domain; that is, it is continuous on its domain.

Proof |

1. Let $P(x)$ be a polynomial function. Since polynomials are formed by adding, subtracting, and multiplying constant functions and the identity function $f(x) = x$, and both of these functions are continuous everywhere, by the Properties of Continuous Functions theorem, $P(x)$ is continuous for all real numbers.
2. Let $R(x) = \frac{P(x)}{Q(x)}$ be a rational function, where $P(x)$ and $Q(x)$ are polynomial functions. Since both $P(x)$ and $Q(x)$ are continuous everywhere (from part 1), and provided that $Q(a) \neq 0$, by the Properties of Continuous Functions theorem, $R(x)$ is continuous at every point in its domain.

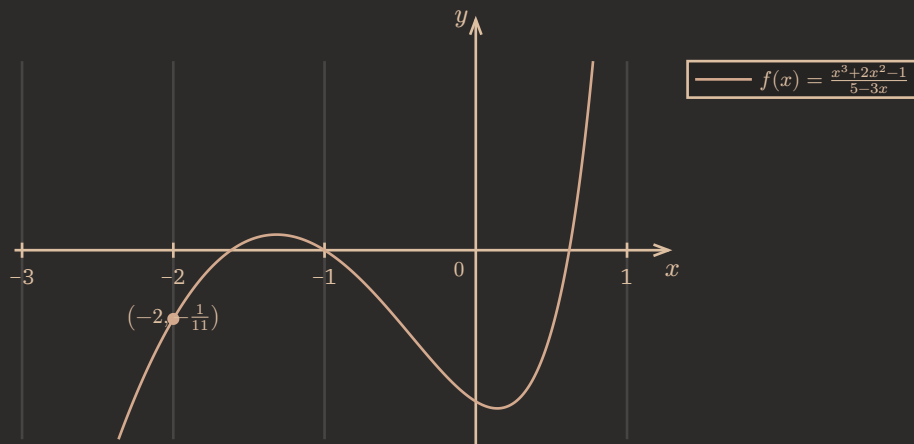
Example | Continuity of a Rational Function

Find $\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x}$.

Solution 1 |

Since the function is a rational function and 2 is in the domain of the function, we can use the Continuity of Rational Functions theorem.

$$\lim_{x \rightarrow -2} \frac{x^3 + 2x^2 - 1}{5 - 3x} = \frac{-2^3 + 2*(-2)^2 - 1}{5 - 3*(-2)} = \frac{-8 + 8 - 1}{5 + 6} = -\frac{1}{11}$$



Example |

Evaluate

$$\lim_{x \rightarrow \pi} \frac{\sin x}{2 + \cos x}$$

Solution 1 |

$\sin x$ and $2 + \cos x$ are continuous at $x = \pi$ and the denom isn't 0 thus continuous. Plugging in,

$$\therefore \frac{0}{3} = 0$$

Are limit and composed function interchangeable?

Example | Counter example

Let $f(x) = H(x)$ (Heaviside), $g(x) = \sin x$

$$f\left(\lim_{x \rightarrow 0} g(x)\right) = f(0) = 1$$

However,

$$\lim_{x \rightarrow 0^-} f(\sin x) = 0, \quad \lim_{x \rightarrow 0^+} f(\sin x) = 1$$

Duh.

Theorem |

If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$ then $\lim_{x \rightarrow a} f(g(x)) = f(b)$.

In other words,

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

Proof |

Let $\varepsilon > 0$. There exists $\delta_1, \delta_2 > 0$ such that

$$\begin{aligned} |y - b| < \delta_1 &\implies |f(y) - f(b)| < \varepsilon, \\ 0 < |x - a| < \delta_2 &\implies |g(x) - b| < \delta_1 \end{aligned}$$

Choose $\delta = \delta_2$. Then for all $\varepsilon > 0$,

$$0 < |x - a| < \delta_2 \implies |f(g(x)) - f(g(a))|$$

3 Intermediate Value Theorem

The Intermediate Value Theorem is a very useful theorem that utilizes the property of continuous functions.

Theorem | Intermediate Value Theorem

Let f be a function that is continuous on the closed interval $[a, b]$, and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then, there exists at least one number c in the open interval (a, b) such that $f(c) = N$.

Equivalently,

Suppose that f is continuous on the closed interval $[a, b]$ and that $f(a)$ and $f(b)$ have opposite signs, i.e., $f(a)f(b) < 0$, then there exists $c \in (a, b)$ such that $f(c) = 0$.

Note | Warning

The Intermediate Value Theorem **does not** tell us how many solutions exists. It guarantees only the existence of the solution.

Proof |

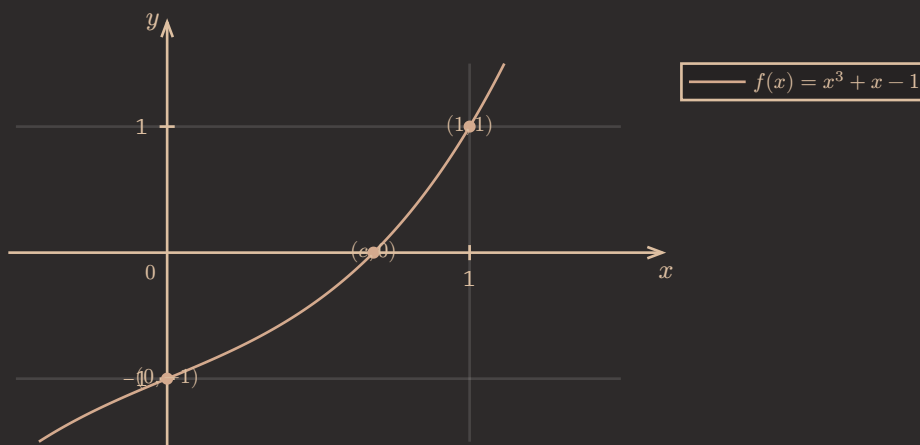
Since f is continuous on the closed interval $[a, b]$, by the definition of continuity, for every x in $[a, b]$, $f(x)$ takes on every value between $f(a)$ and $f(b)$. Therefore, for any number N between $f(a)$ and $f(b)$, there must exist at least one number c in the open interval (a, b) such that $f(c) = N$.

Example | Intermediate Value Theorem Example

Show that the equation $x^3 + x - 1 = 0$ has a solution in the interval $[0, 1]$.

Solution 1 |

Let $f(x) = x^3 + x - 1$. We need to show that there exists a number c in the interval $(0, 1)$ such that $f(c) = 0$. First, we check the values of f at the endpoints of the interval: $f(0) = 0^3 + 0 - 1 = -1$ $f(1) = 1^3 + 1 - 1 = 1$ Since $f(0) = -1$ and $f(1) = 1$, and 0 is between -1 and 1 , by the Intermediate Value Theorem, there exists at least one number c in the interval $(0, 1)$ such that $f(c) = 0$. Therefore, the equation $x^3 + x - 1 = 0$ has a solution in the interval $[0, 1]$.



Chapter 01

Deriveratives

This chapter formalizes the concept of the limit, moving from intuitive approximations to rigorous mathematical definitions including the precise Epsilon-Delta definition. It provides the algebraic rules (Limit Laws) necessary to calculate limits without graphing, methods for handling indeterminate forms, and defines continuity to describe functions without breaks or holes. These concepts culminate in the Intermediate Value Theorem, a powerful tool used to prove the existence of roots and solutions within specific intervals, setting the stage for the definition of the derivative.

Chapter 01.01

Derivatives and Rates of Change

In Chapter 1 we defined limits and learned techniques for computing them. We now revisit the problems of finding tangent lines and velocities from Section 1.4. The special type of limit that occurs in both of these problems is called a derivative and we will see that it can be interpreted as a rate of change in any of the natural or social sciences or engineering.

1 Tangents

Definition |

The derivative of a function f at a number a , denoted by $f'(a)$ is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

Provided that the limit exists.

Example |

Find the derivative of the function $f(x) = x^2 - 8x + 9$ at the number 2.

Solution 1 |

$$\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2} = \lim_{x \rightarrow 2} \frac{x^2 - 8x + 12}{x - 2} = -4$$

Example |

Determine whether $f'(0)$ exists.

$$f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

$$f(x) = \begin{cases} x^2 \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Solution 1 |

$$\frac{f(x) - f(0)}{x - 0} = \frac{x \sin\left(\frac{1}{x}\right) - 0}{x - 0} = \sin \frac{1}{x}, \quad x \neq 0$$

We don't know the existence so we put limit afterwards. Since $\lim_{x \rightarrow 0} \sin \frac{1}{x}$ does not exist, f is not differentiable at 0.

Solution 2 |

$$\frac{f(x) - f(0)}{x - 0} = \frac{x^2 \sin\left(\frac{1}{x}\right) - 0}{x - 0} = x \sin \frac{1}{x}, \quad x \neq 0$$

Thus,

$$f'(0) = \lim_{x \rightarrow 0} x \sin \frac{1}{x} = 0$$

Squeeze Theorem.

Definition |

The **tangent line** to the curve $y = f(x)$ at the point $P(x, f(x))$ is line through P such that the slope is $f'(x)$.

Example |

Find an equation of the tangent line to the hyperbola $y = \frac{3}{x}$ at $(3, 1)$.

Solution 1 |

$$f'(3) = \lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3} = \lim_{x \rightarrow 3} \frac{\frac{3}{x} - 1}{x - 3} = \lim_{x \rightarrow 3} \frac{3 - x}{x(x - 3)} = \lim_{x \rightarrow 3} -\frac{1}{x} = -\frac{1}{3}$$

So the equation of the tangent line is $y - 1 = -\frac{1}{3}(x - 3)$, or $y = -\frac{1}{3}x + 2$.

2 Velocities

3 Derivatives

4 Rates of Change

Definition |

Let $y = f(x)$. If x changes from x_1 to x_2 , then the **change (or increment, 증분)** in x is

$$\Delta x = x_2 - x_1$$

and the corresponding increment in y is

$$\Delta y = f(x_2) - f(x_1)$$

The difference quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

is called the average rate of change.

Definition |

Taking the limit as $\Delta x \rightarrow 0$ gives the **(instantaneous) rate of change**(순간변화율) of y with respect to x at $x = x_1$. That is the derivative:

$$f'(x) = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

Example | Symmetric difference quotient

Suppose f is differentiable at a . Then

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a-h)}{2h} = f'(a)$$

Intuitively.

Proof |

$$\begin{aligned}
& \lim_{h \rightarrow 0} \frac{f(a+h) - f(a-h)}{2h} \\
&= \frac{1}{2} \left[\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} + \lim_{h \rightarrow 0} \frac{f(a) - f(a-h)}{h} \right] \\
&= \frac{f'(a) + f'(a)}{2} = f'(a)
\end{aligned}$$

Note | Remark

The symmetric difference quotient is known to converge faster than usual difference quotient. Moreover, the converse of the statement does not hold in general. This is not a definition, so the differentiability on a may be required.

Example | Counterexample

$$\begin{aligned}
f(x) &= |x| \\
f'(0) &=?
\end{aligned}$$

Example | Exercise

Let f be differentiable at a .

$$\lim_{h \rightarrow 0} g(ch) = \lim_{t \rightarrow 0} g(t)$$

Example | Exercise

Let f be differentiable at a .

$$\lim_{h \rightarrow a} \frac{f(a) - f(a-h)}{h} = f'(a)$$

Chapter 01.02

The Derivative as a Function

Definition |

Recall that the derivative of f at a number a is defined (when the limit exists)

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Since a is arbitrary, we may replace a by x and obtain a new function

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{t \rightarrow x} \frac{f(t) - f(x)}{t - x}$$

called the **derivative** of f .

To plot a graph of $f'(x)$, estimate the slope of the tangent line to the graph of f at several values of x .

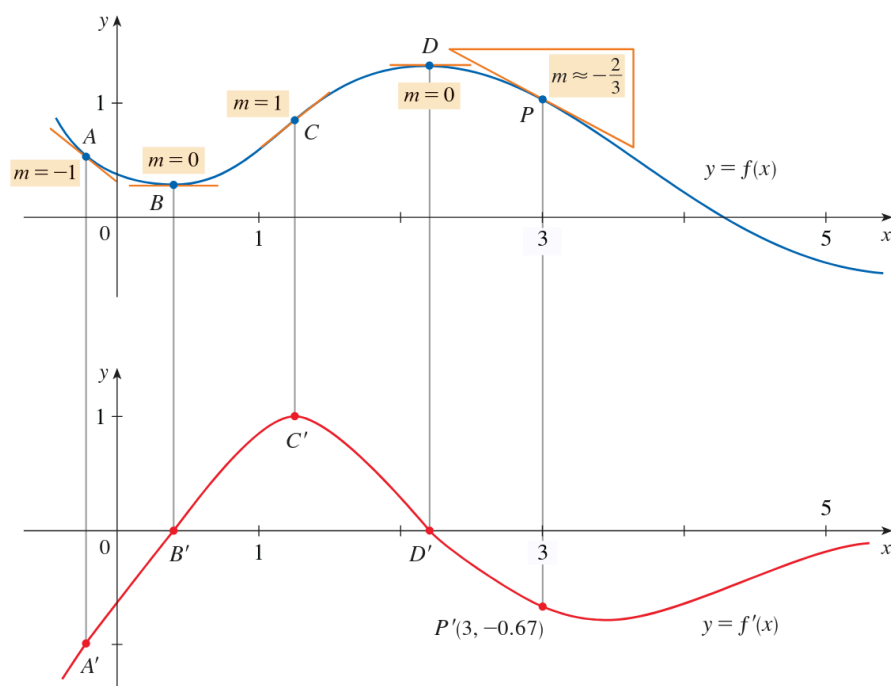


FIGURE 2

Example |

If $f(x) = \sqrt{x}$, find the derivative of f . State the domain of f'

Solution 1 |

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h} + \sqrt{x}} = \frac{1}{2\sqrt{x}} \end{aligned}$$

The domain has changed.

Definition |

A function f is **differentiable** at a if $f'(a)$ exists. It is differentiable on an open interval (a, b) if it is differentiable at every number in the interval.

Notation |

If $y = f(x)$, the derivative of f may be written as:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx}f(x) = Df(x) = D_x f(x)$$

The symbols D and $\frac{d}{dx}$ are called **differentiation operators**, and they denote the operation of differentiation.

Note |

Usually, the term *operator* is regarded as a higher order function, that takes another functions as input or output.

Note | Warning

The notation $\frac{dy}{dx}$ introduced by Leibniz, ***should not be regarded as a ratio***. It is simply another notation for $f'(x)$

Notation |

In Leibniz notation, the value of the derivative at $x = a$, we write

$$\frac{dy}{dx}\bigg|_{x=a} \quad \frac{dy}{dx}\bigg]_{x=a}$$

Theorem |

If f is differentiable at a , then f is continuous at a .

Proof |

$$\begin{aligned}\lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} \left[\frac{f(x) - f(a)}{x - a} (x - a) + f(a) \right] \\ &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a) + f(a) \\ &= f'(a) \cdot 0 + f(a) = f(a)\end{aligned}$$

The converse of the theorem is false. $|x|$ is an example.

Weierstrass function. Continuous everywhere but differentiable nowhere.

Definition |

The **Second derivative**(이계도함수) of f is the derivative of f' :

$$f''(x) = (f')'(x) = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d^2 y}{dx^2}$$

The **Third derivative** of f is the derivative of f'' :

$$f'''(x) = (f'')'(x) = \frac{d}{dx} \left(\frac{d}{dx} \left(\frac{dy}{dx} \right) \right) = \frac{d^3 y}{dx^3}$$

The **n th derivative** of f is denoted by

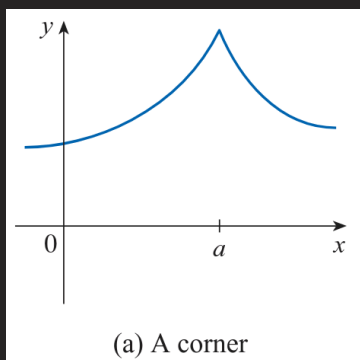
$$f^{(n)}(x) = \frac{d^n y}{dx^n}$$

If the function denotes the position of an object for given time, its derivative is velocity, second derivative is acceleration, and third derivative is *jerk*(?).

1 How can a function fail to be differentiable?

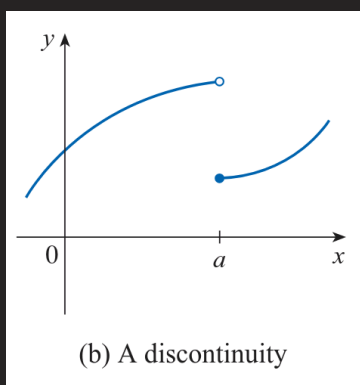
1.1 Corner of kink

If the graph of f has a corner or kink at $x = a$, then the tangent line do not exist.



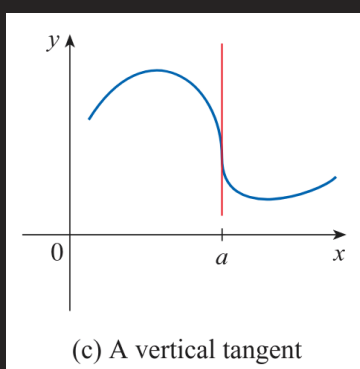
1.2 Discontinuity.

If f is not continuous at $x = a$, then the tangent line do not exist.



1.3 Vertical tangent

If the tangent line is vertical, its slope does not exist.



2 One-Sided Derivatives

Definition |

The **left-handed derivatives** of f at a point is defined by

$$f'_-(a) = \lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}$$

Simillar to right handed derivatives.

Chapter 01.03

Differentiation Formulas

1 Constant Functions and Power Functions

1. If $f(x) = c$, then $f'(x) = 0$
2. If $f(x) = x^n$ ($n \in \mathbb{N}$), then $f'(x) = nx^{n-1}$

Proof |

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{c - c}{h} = 0 \end{aligned}$$

Proof |

$$\begin{aligned} f'(x) &= \lim_{t \rightarrow x} \frac{f(x) - f(t)}{x - t} \\ &= \lim_{t \rightarrow x} \frac{x^n - t^n}{x - t} \\ &= \lim_{t \rightarrow x} \frac{(x - t)(x^{n-1} + x^{n-2}t + \dots + xt^{n-2} + t^{n-1})}{x - t} \\ &= \lim_{t \rightarrow x} (x^{n-1} + x^{n-2}t + \dots + xt^{n-2} + t^{n-1}) \\ &= x^{n-1} + x^{n-1} + \dots + x^{n-1} \quad (\text{n terms}) \\ &= nx^{n-1} \end{aligned}$$

2 Derivative Rules

Definition |

1. $(f \pm g)'(x) = f'(x) \pm g'(x)$
2. $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$
3. $(cf)'(x) = cf'(x)$
4. $\left(\frac{f(x)}{g(x)}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$

Proof |

Let $F(x) = f(x) + g(x)$

$$\begin{aligned}
 F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h} \\
 &= \lim_{h \rightarrow 0} \left[\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \right] \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h} \\
 &= f'(x) + g'(x)
 \end{aligned}$$

Proof |

Let $F(x) = f(x)g(x)$. Add and subtract $f(x)g(x+h)$ in the numerator:

Since $g(x)$ is differentiable, $g(x)$ is continuous.

$$\begin{aligned}
 F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h)g(x+h) - f(x)g(x+h) + f(x)g(x+h) - f(x)g(x)}{h} \\
 &= \lim_{h \rightarrow 0} \left[\left(\frac{f(x+h) - f(x)}{h} \right) g(x+h) + f(x) \left(\frac{g(x+h) - g(x)}{h} \right) \right] \\
 &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \cdot \lim_{h \rightarrow 0} g(x+h) + f(x) \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} \\
 &= f'(x)g(x) + f(x)g'(x)
 \end{aligned}$$

Proof |

Let $F(x) = \frac{f(x)}{g(x)}$.

$$\begin{aligned}
F'(x) &= \lim_{h \rightarrow 0} \frac{F(x+h) - F(x)}{h} \\
&= \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)} \right) \\
&= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \\
&= \lim_{h \rightarrow 0} \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \\
&= \lim_{h \rightarrow 0} \frac{\left(\frac{f(x+h)-f(x)}{h} \right) g(x) - f(x) \left(\frac{g(x+h)-g(x)}{h} \right)}{g(x+h)g(x)} \\
&= \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}
\end{aligned}$$

Example |

1. $x^3 + 7x^2 - 3x + 12$
2. $(1 + 2x^2)(x - x^2)$
3. $\frac{x^2+1}{x-1}$

Solution 1 |

$$\begin{aligned}
y' &= (x^3 + 7x^2 - 3x + 12)' = (x^3)' + (7x^2)' - (3x)' + (12)' \\
&= 3x^2 + 14x - 3
\end{aligned}$$

Solution 2 |

$$\begin{aligned}
y' &= (1 + 2x^2)'(x - x^2) + (1 + 2x^2)(x - x^2)' \\
&= 4x(x - x^2) + (1 + 2x^2)(1 - 2x) \\
&= -4x^3 + 4x^2 - 4x^3 + 2x^2 - 2x + 1 \\
&= -8x^3 + 6x^2 - 2x + 1
\end{aligned}$$

Or expand and do it like sol 1

Solution 3 |

Using the quotient rule:

$$\begin{aligned}y' &= \frac{((x^2 + 1))'(x - 1) - (x^2 + 1)((x - 1))'}{(x - 1)^2} \\&= \frac{2x(x - 1) - (x^2 + 1)(1)}{(x - 1)^2} \\&= \frac{2x^2 - 2x - x^2 - 1}{(x - 1)^2} \\&= \frac{x^2 - 2x - 1}{(x - 1)^2}\end{aligned}$$

3 General Power Functions

Theorem |

If $f(x) = x^{-n}$ ($n \in \mathbb{N}$), then $f'(x) = -nx^{-n-1}$

Proof |

Rewrite the function as a fraction: $f(x) = \frac{1}{x^n}$.

Now, apply the quotient rule:

$$\begin{aligned}f'(x) &= \frac{(1)'(x^n) - (1)(x^n)'}{(x^n)^2} \\&= \frac{0 \cdot x^n - nx^{n-1}}{x^{2n}} \\&= \frac{-nx^{n-1}}{x^{2n}} \\&= -nx^{n-1-2n} \\&= -nx^{-n-1}\end{aligned}$$

We will later prove the general power rule for real exponents. For now we state it and will use it in examples and exercises.

Theorem |

If $f(x) = x^r$ ($x \in \mathbb{R}$), then $f'(x) = rx^{r-1}$

Example |**Reciprocal Rule****Example | Tangent lines**

Find the equations of the tangent line and normal line to the curve $y = \frac{\sqrt{x}}{1+x^2}$ at the point $(1, \frac{1}{2})$.

Solution 1 |

$$y' = \frac{(\sqrt{x})'(1+x^2) - \sqrt{x}(1+x^2)'}{(1+x^2)^2} = \frac{\frac{1}{2\sqrt{x}}(1+x^2) - \sqrt{x}(2x)}{(1+x^2)^2}$$

Thus the slope at the point is $-\frac{1}{4}$

$$\therefore y = -\frac{1}{4}x + \frac{3}{4} \quad y = 4x - \frac{7}{2}$$

Chapter 01.04

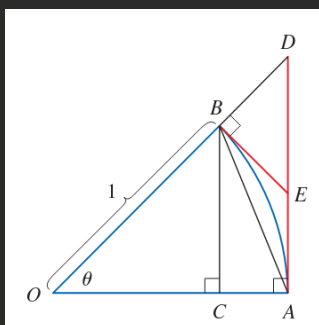
Derivatives of Trigonometric Functions

1 Three Special Trigonometric Limits

Theorem | Three Special Trigonometric Limits

1. $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
2. $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$
3. $\lim_{x \rightarrow 0} \frac{\tan x}{x} = 1$

Proof |



$$|BC| = \sin \theta < \widehat{AB} = \theta < |AE| + |BE| < |AE| + |DE| = \tan \theta$$

$$\implies \sin \theta < \theta < \tan \theta \iff \cos \theta < \frac{\sin \theta}{\theta} < 1$$

Since $0 < \sin \theta < \theta$

Using the squeeze theorem,

$$\lim_{\theta \rightarrow 0^+} \sin \theta = 0$$

To evaluate the limit as $\theta \rightarrow 0^-$ let $\theta = -t$. Then $\theta \rightarrow 0^+$ and $\sin$ is odd function. Hence

Proof |

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} =$$

Example |

Find

$$\lim_{x \rightarrow 0} \frac{\sin 7x}{4x}$$

Solution 1 |

$$\lim_{x \rightarrow 0} \frac{\sin 7x}{4x} = \lim_{x \rightarrow 0} \frac{\sin 7x}{7x} \cdot \frac{7}{4} = \frac{7}{4}$$

Can we just substitute it??

Example |

$$\lim_{x \rightarrow 0} \csc x \sin(\sin x)$$

Solution 1 |

$$= \lim_{x \rightarrow 0} \frac{\sin(\sin x)}{\sin x}$$

Let

$$f(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

f is continuous.

$$= \lim_{x \rightarrow 0} \frac{\sin(\sin x)}{\sin x} = \lim_{x \rightarrow 0} f(\sin x) = f\left(\lim_{x \rightarrow 0} \sin x\right) = f(0) = 1$$

2 Derivatives of Trig Functions

Theorem |

1. $\frac{d}{dx} \sin x = \cos x$
2. $\frac{d}{dx} \cos x = -\sin x$
3. $\frac{d}{dx} \tan x = \sec^2 x$
4. $\frac{d}{dx} \csc x = -\csc x \cot x$
5. $\frac{d}{dx} \sec x = \tan x \sec x$
6. $\frac{d}{dx} \cot x = -\csc^2 x$

Proof |

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequaleam animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

Note | Remark

The derivatives of $\sin x$ and $\cos x$ repeat in a cycle of length 4.

$$\sin x \rightarrow \cos x \rightarrow -\sin x \rightarrow -\cos x \rightarrow \sin x$$

$$\cos x \rightarrow -\sin x \rightarrow -\cos x \rightarrow \sin x \rightarrow \cos x$$

Example | no mod

Derivatives of $\cos x$ repeat in a cycle of length 4. Since $27 = 4 \cdot 6 + 3$,

$$\cos^{27} x = \cos^3 x = \sin x$$

Do not use modulo here

Chapter 01.05

The Chain Rule

Definition |

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and

$$\begin{aligned} F'(x) &= f'(g(x)) \cdot g'(x) \\ &= \frac{d}{dg} \cdot \frac{dg}{dx} \end{aligned}$$

Proof |

$$\begin{aligned} \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a} \\ &= \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{f(x) - g(x)} \cdot \frac{f(x) - g(x)}{x - a} \\ &= \lim_{x \rightarrow a} f'(g(a)) \cdot g'(a) \end{aligned}$$

Is it substitutable? what if $f(x)$ - $g(x)$ approach 0?

Proof |

Define function $\varepsilon(x)$ which denotes error, that is continuous, avoiding fraction notation.

Example |

Using the Chain Rule prove followings:

- Derivatives of odd functions are even and vice versa.

Chapter 01.06

Implicit Differentiation

1 Implicit Function

Definition |

Equation involving x and y of the form $F(x, y) = 0$

Is called the equation in form of **implicit function**(음함수 형태).

2 Implicit Function Theory

Theorem |

In many cases an equation of the form $F(x, y) = 0$ determines y as a function of x near a given point.

Example |

$$x^2 + y^2 = 25$$

For points $y > 0$,

$$y = \sqrt{25 - x^2}$$

For points $y < 0$,

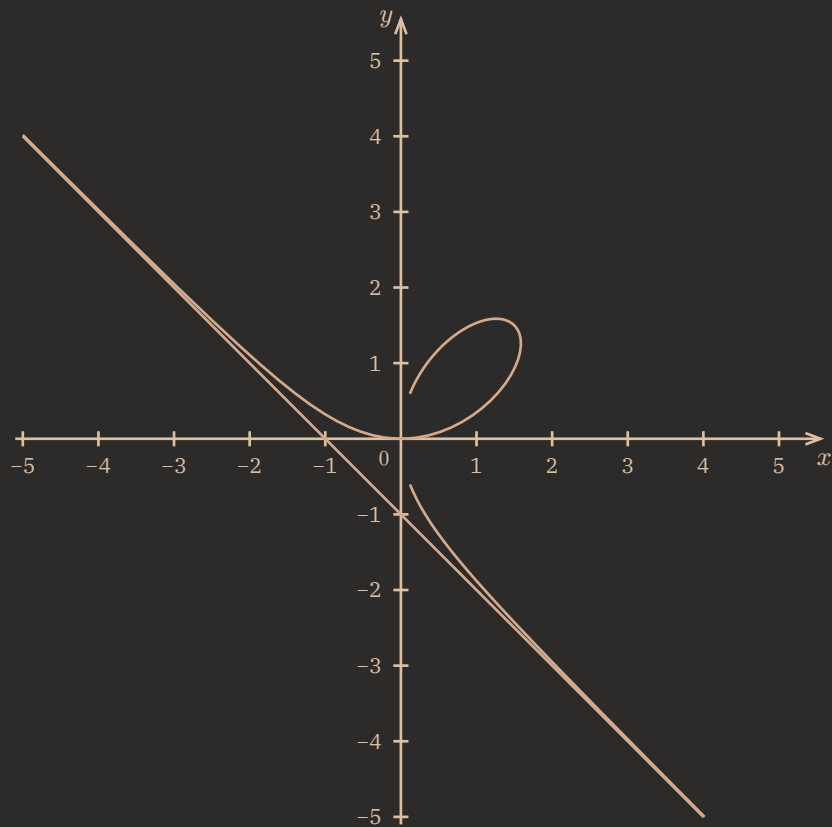
$$y = -\sqrt{25 - x^2}$$

However, on point $(5, 0)$, $(-5, 0)$ we can't express y as a function.

In some cases, it is difficult to solve an equation explicitly for y as a function of x . For example, the folium of Descartes

Example |

$$x^3 + y^3 = 6xy$$



By the implicit function theory, we can express it as a function on near the point.
Thus, we can treat $y = f(x)$

Example |

$$x^2 + y^2 = 25$$

Solution 1 |

$$x^2 + (f(x))^2 = 25$$

Differentiate both side

$$\frac{d}{dx}(x^2 + (f(x))^2) = \frac{d}{dx}25$$

$$2x + 2f(x) \cdot f'(x) = 0$$

$$f'(x) = -\frac{x}{f(x)} = -\frac{x}{y}$$

Example |

Find y' of the Descartes foillum

Solution 1 |

$$y^3 + x^3 = 6xy$$

$$\frac{d}{dx}(y^3 + x^3) = \frac{d}{dx}(6xy)$$

$$\frac{d}{dx}(y^3) + \frac{d}{dx}(x^3) = 6y$$

$$3y^2 \frac{dy}{dx} + 3x^2 = 6y$$

$$\frac{dy}{dx} = \frac{6y - 3x^2}{3y^2} \quad (y \neq 0)$$

3 Second Derivatives of IMplicit Functions

Example |

Find $\frac{d^2}{dx^2}y$ for

$$x^4 + y^4 = 16$$

Solution 1 |

Do it twice.

$$\frac{d}{dx}(x^4 + y^4) = \frac{d}{dx}16$$

$$4x^3 + 4y^3 \frac{dy}{dx} = 0$$

Chapter 01.07

Rates of Change in the Natural and Social Sciences

woiefj

Chapter 01.08

Relative Rates

Sometimes rate of the variable is related to other variable's rate.

Example |

Car A is traveling west at 50 mi/h and car B is traveling north at 60 mi/h. Both are headed for the intersection of the two roads. At what rate are the cars approaching each other when car A is 0.3 mi and car B is 0.4 mi from the intersection?

Solution 1 |

At given time t , let x be the distance from car A to intersection, let y be the distance from car B to intersection. Then, by the Pythagorean theorem, distance d is

$$d^2 = x^2 + y^2$$

Implicit differentiation both side

$$2z \frac{dz}{dt} = 2x \frac{dx}{dt} + 2y \frac{dy}{dt}$$

At $x = 0.3$, $y = 0.4$, then $z = 0.5$ and

$$\frac{dz}{dt} = \frac{1}{0.5} [0.3(-50) + 0.4(-60)] = -78$$

The cars are approaching each other at a rate of -78 mi/h

Example |

Chapter 01.09

Linear Approximations and Differentials

When x is close to a , the tangent line of f at $x = a$

Notation |

The approximation $f(x) \approx L(x)$, or equivalently,

$$f(x) \approx f(a) + f'(a)(x - a)$$

Theorem |

Let $y = f(x)$ and it is a differentiable function. To describe how the value of f changes by change of x , we use **differentials**

Consider dx as some sort of rate of change of x , and respectively for y . Then,

$$dy = f'(x) dx$$

Somewhat like a generalization of **derivative**... we can't define precisely now.

Chapter 02

Applications of Differentiation

This chapter formalizes the concept of the limit, moving from intuitive approximations to rigorous mathematical definitions including the precise Epsilon-Delta definition. It provides the algebraic rules (Limit Laws) necessary to calculate limits without graphing, methods for handling indeterminate forms, and defines continuity to describe functions without breaks or holes. These concepts culminate in the Intermediate Value Theorem, a powerful tool used to prove the existence of roots and solutions within specific intervals, setting the stage for the definition of the derivative.

Chapter 02.01

Maximum and Minimum Values

Definition |

Let c be a number in the domain D of a function f . Then $f(c)$ is the

- (absolute, global) **maximum value** of f on D if $f(c) \geq f(x)$ for all $x \in D$
- (absolute, global) **minimum value** of f on D if $f(c) \leq f(x)$ for all $x \in D$

These are called **extreme values** of f .

Definition |

The number $f(c)$ is a

- **local maximum value** of f if $f(c) \geq f(x)$ when x is “near” c .
- **local minimum value** of f if $f(c) \leq f(x)$ when x is “near” c .

We say that something is *near* c , we mean that it is true on some open interval containing c . In particular, every global maximum is local maximum and vice versa.

Note |

The book somehow do not consider the endpoints of the interval as extrema. i.e., the domain is open interval. Thus if the function value at endpoint is smaller/greater than any other, the function do not have extreme value.

Theorem | The Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$ then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some $c, d \in [a, b]$.

Theorem | Fermat's Theorem

If f has a local minimum or maximum at c and if $f'(c)$ exists, $f'(c) = 0$

Proof |

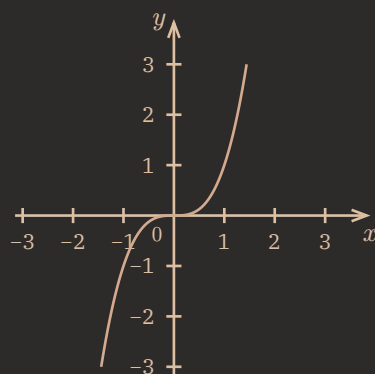
Suppose f has a local minimum at c . Then for h close to 0,

$$f(c + h) - f(c) \geq 0$$

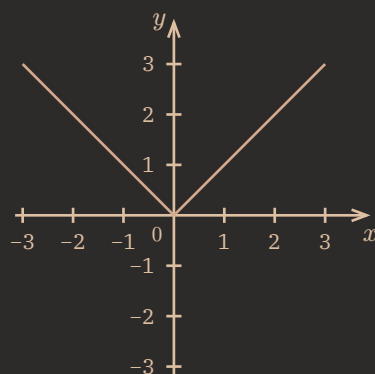
If $h > 0$,

Note |

The converse of Fermat's theorem is not true.



at $x = 0$, $f'(0) = 0$ but not extreme value. Also, Fermat's Theorem is only valid for when it is differentiable.



at $x = 0$, $f'(0)$ doesn't exist and is minimum value. Therefore, you should check the differentiable interval before finding critical values.

Definition | Critical Numbers

Points where it can potentially be maxima / minima.

1. Points where $f'(x) = 0$
2. Points where $f'(x)$ is not defined.
- 3.

Theorem | Closed Interval Method

Investigate all candidates.

1. critical numbers
2. endpoints of the interval

The largest of the values from 1 and 2 is absolute minimum/maximum value.

Chapter 02.02

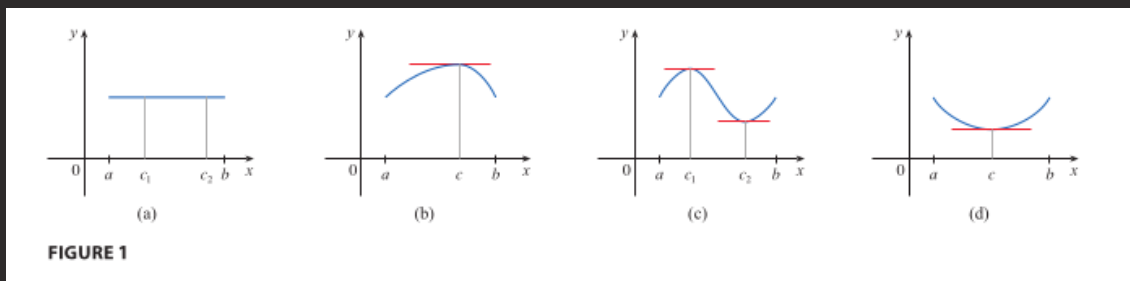
The Mean Value Theorem

Theorem | Rolle's Theorem

Let f be a function that satisfies the following three hypothesis:

1. f is continuous on the closed interval $[a, b]$
2. f is differentiable on the open interval (a, b)
3. $f(a) = f(b)$

Then there exists a number $c \in (a, b)$ such that $f'(c) = 0$

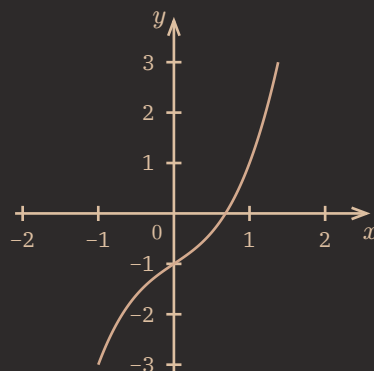


Proof |

1. Case: $f(x) = k$, a constant function
2. Case: $f(x) > f(a)$ for some $x \in (a, b)$
3. Case: $f(x) < f(a)$ for some $x \in (a, b)$

Example |

Prove that the equation $x^3 + x - 1 = 0$ has exactly one real solution.



Solution 1 |

First we use IVT to show that a solution exists. Let $f(x) = x^3 + x - 1$.

$$(f(0) < 0) \wedge (f(1) > 0) \implies \exists c \in (0, 1) \text{ such that } f(c) = 0$$

By Rolle's Theorem, we can show uniqueness. (TODO)

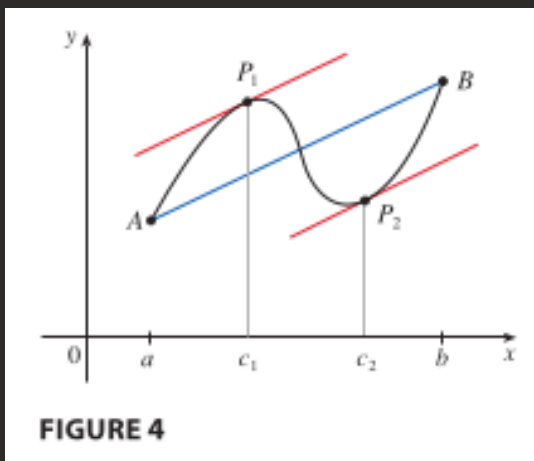
Theorem |

Let f be a function that satisfies the followings

1. f is continuous on the closed interval $[a, b]$
2. f is differentiable on the open interval (a, b)

Then there is a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$



Proof |

$$h(x) = f(x) - \left[\frac{f(b) - f(a)}{b - a} (x - a) + f(a) \right]$$

Then h is continuous and differentiable.

$$h(a) = f(a) - f(a) = 0$$

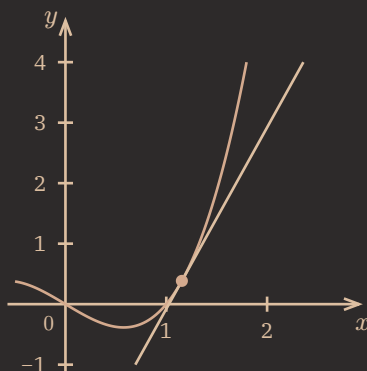
$$h(b) = f(b) - f(b) = 0$$

Using Rolle's Theorem, proven!

Example |

$$f(x) = x^3 - x$$

Find $c \in (0, 2)$ satisfying the MVT.

**Solution 1 |**

$$f'(c) = \frac{f(2) - f(0)}{2 - 0} = 3$$

$$f'(x) = 3x^2 - 1$$

$$c = \sqrt{\frac{4}{3}}$$

Example |

Suppose that $f(0) = -3$ and $f'(x) \leq 5$ for all values of x . How large can $f(2)$ possible be?

Solution 1 |

By MVT

$$\exists c \in (0, 2) \quad f(2) - f(0) = f'(c)(2 - 0)$$

$$f(2) - f(0) \leq 10$$

$$f(2) \leq 7$$

Sharpness: Let $f(x) = 5x - 3$. Then $f(2) = 7$

Example |

Show that

$$\sin x < x \text{ if } 0 < x < 2\pi$$

Theorem |

If $f'(x) = 0$ for all x in an interval (a, b) , then f is constant of (a, b)

Proof |

Let $x_1, x_2 \in (a, b)$ with $f(x_1) \neq f(x_2) \wedge x_1 < x_2$. Since f is differentiable on (a, b) . By MVT

$$\exists c \in (x_1, x_2) \quad f(x_2) - f(x_1) = (x_2 - x_1)f'(c)$$

Contradiction.

Theorem |

If $f'(x) = g'(x)$, there exists c such that $f(x) = g(x) + c$

Proof |

Let $F(x) = f(x) - g(x)$

$$F'(x) = f'(x) - g'(x)$$

What Derivatives Tell Us about the Shape of a Graph

Since $f'(x)$ represents the slope of the curve $y = f(x)$ at the point $(x, f(x))$, it tells us the direction in which the curve proceeds at each point.

1 I/D Test

Theorem | Increasing/Decreasing Test (I/D Test)

- (a) If $f'(x) > 0$ on an interval, then f is increasing on that interval
- (b) If $f'(x) < 0$ on an interval, then f is decreasing on that interval

$f'(x) = 0$ needs caution.

Note |

$$\begin{cases} a < b \implies f(a) < f(b) & \text{(strictly) increasing} \\ a < b \implies f(a) \leq f(b) & \text{(monotonically) increasing} \end{cases}$$

Example |

Find where the function

$$f(x) = 3x^4 - 4x^3 - 12x^2 + 5$$

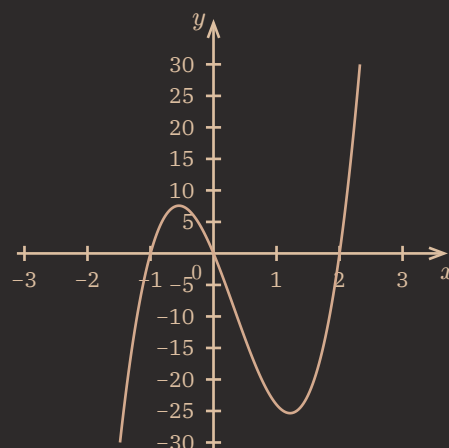
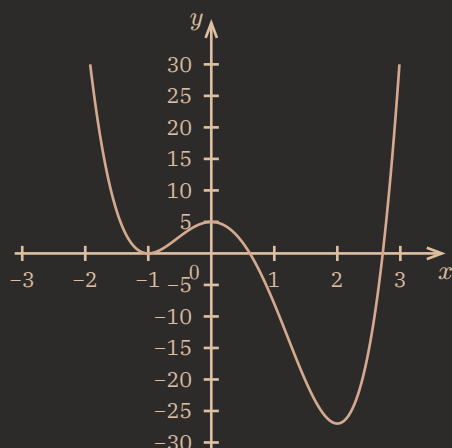
is increasing or decreasing.

Solution 1 |

$$f'(x) = 12x^3 - 12x^2 - 24x = 12x(x^2 - x - 2) = 12x(x + 1)(x - 2)$$

x	-1	...
$f(x)$	0	-
$f'(x)$	?	?

Differentiate to figure out the shape of the graph.



Each $f(x)$ and $f'(x)$, respectively.

2 First Derivative Test

The *First Derivative Test* is a consequence of the *I/D Test*.

Theorem | First Derivative Test

Suppose that c is a critical number of a continuous function f .

- (a) If f' changes from positive to negative at c , then f has a local maximum at c .
- (b) If f' changes from negative to positive at c , then f has a local minimum at c .
- (c) If f' is positive to the left and right of c , or negative to the left and right of c , then f has no local maximum or minimum at c .

Example |

Find the local maximum and minimum values of the function

$$f(x) = x + 2 \sin x, \quad 0 \leq x \leq 2\pi$$

Solution 1 |

3 Concavity

Definition | Concavity

If the graph of f lies above all of its tangents on an interval I , then f is called **concave upward** (위로 오목, 아래로 볼록) on I . If the graph of f lies below all of its tangents on I , then f is called **concave downward** (아래로 오목, 위로 볼록) on I .

Note |

This *concavity* notation is not conventional, we usually use *convexity*. However, since Stewart Textbook is using concavity notation, we follow the prior.

Definition | Inflection Point

A point P on a curve $y = f(x)$ is called an **inflection point** if f is continuous there and the curve changes from concave upward to concave downward or vice versa at P .

Theorem | Concavity Test

- (a) If $f''(x) > 0$ on an interval I , then the graph of f is concave upward on I .
- (b) If $f''(x) < 0$ on an interval I , then the graph of f is concave downward on I .

Note |

If you are confused, think of the function $y = x^2$, which is definitely concave upward.

Theorem | The Second Derivative Test

Suppose f'' is continuous near c .

- (a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c .
- (b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c .

Example |

Sketch the graph of

$$f(x) = x^{\frac{2}{3}}(6 - x)^{\frac{1}{3}}$$

Solution 1 |

$$f'(x) = \frac{2}{3}x^{-\frac{1}{3}}(6 - x)^{\frac{1}{3}} - \frac{1}{3}x^{\frac{2}{3}}(6 - x)^{-\frac{2}{3}}$$

Note |

Chapter 02.04

Limits at infinity; Horizontal Asymptotes

In this section, we consider what happens to y as x becomes arbitrarily large, either positively or negatively. This so-called **end behavior** is very useful when sketching graphs.

Definition |

$$\lim_{x \rightarrow \infty} f(x) = L \left(\text{resp. } \lim_{x \rightarrow -\infty} f(x) = \right)$$

Theorem |

If $r > 0$ is a rational number, then

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0$$

If $r < 0$ is that

Example |

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1}$$

Solution 1 |

$$\lim_{x \rightarrow \infty} \frac{3x^2 - x - 2}{5x^2 + 4x} = \lim_{x \rightarrow \infty}$$

1 Precise Definition of a Limit at Infinity

Definition |

Let f be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L$$

means that from every $\varepsilon > 0$, there is a corresponding number N such that

$$x > N \implies |f(x) - L| < \varepsilon$$

Theorem |

$$\lim_{x \rightarrow \infty} f(x) = \infty \quad \lim_{x \rightarrow \infty} g(x) = L$$

$$1. \lim_{x \rightarrow \infty} [f(x) + g(x)] = \infty$$

Example |

Show that $\lim_{x \rightarrow \infty} \sin x$ does not exist

Solution 1 |

Suppose that $\lim_{x \rightarrow \infty} \sin x = L$ for some

Example |

Modify the above argument to show that the Dirichlet function

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

So that it is not continuous on 0.

Solution 1 |

Suppose f is continuous on $a \in \mathbb{Q}$

$$|x - a| < \delta_1 \implies |f(x) - f(a)| < \frac{1}{2}$$

$$\frac{1}{2} < f(x) < \frac{3}{2}$$

It must all be rational number in range $(a - \delta_1, a + \delta_1)$. Contradiction.

Example |

Find the limit or show that it does not exist.

$$\lim_{x \rightarrow \infty} \sqrt{x} \sin \frac{1}{x}$$

Solution 1 |

$\lim_{x \rightarrow \infty}$ implies $t \rightarrow 0^+$ when $t = \frac{1}{x}$.

Note |

Always check **Periodicity**, **Symmetry**, **Domain** of the function.

Chapter 02.05

Summary of Curve Sketching

Note |

Always check **Periodicity**, **Symmetry**, **Domain** of the function.

Note |

1. Intervals of Increase and Decrease
2. End Behaviour
 1. Asymptotes (Horizontal, Slant)
3. Horizontal Asymptotes
4. Concavity
5. Intervals of Increase or Decrease
6. Local Maximum or Minimum Values
7. Points of Inflection
8. Domain
9. Intercepts
10. Symmetry
11. $\vdots$

Definition |

f has a slant asymptote of $mx + b$ when

$$\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0$$

Theorem |

If

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = m$$

$$\lim_{x \rightarrow \infty} (f(x) - mx) = b$$

Then,

$$\lim_{x \rightarrow \infty} [f(x) - (mx + b)] = 0$$

Example |

Find the slant asymptote of $f(x) = \sqrt{4x^2 + 4x}$

Solution 1 |

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \sqrt{4 + \frac{4}{x}} = 2$$

$$\lim_{x \rightarrow \infty} [f(x) - 2x] = \lim_{x \rightarrow \infty} \frac{(4x^2 + 4x) - 4x^2}{\sqrt{4x^2 + 4x} + 2x} = 1$$

Therefore, it has a slant asymptote $y = 2x + 1$.

Example |

Use the guidelines to sketch the curve

$$y = \frac{2x^2}{x^2 - 1}$$

Solution 1 |

Domain: $\mathbb{R} \setminus \{-1, 1\}$

Intercepts: x-intercepts 0, y-intercepts 0

Symmetry: Even function.

Periodicity: any nonconstant rational function is not a periodic function.

Asymptote:

Horizontal asymptotes.

$$\lim_{x \rightarrow \infty} \frac{2x^2}{x^2 - 1} = 2 \implies y = 2$$

Since f is even,

$$\lim_{x \rightarrow -\infty} f(x) = 2$$

Thus $y = 2$ is the only horizontal asymptote.

Alternatively, $f(x) = 2 + \frac{2}{x^2-1}$

Since there is a horizontal asymptote, obviously, slant asymptote doesn't exist.

Vertical Asymptotes

For $x \neq 1$, f is continuous. Thus the only possible candidates are $x = \pm 1$.

$$\lim_{x \rightarrow -1^-} f(x) = -\infty \quad \lim_{x \rightarrow -1^+} f(x) = \infty$$

$$\lim_{x \rightarrow 1^-} f(x) = -\infty \quad \lim_{x \rightarrow 1^+} f(x) = \infty$$

Intervals of Increase or Decrease, Local Maximum or Minimum Values, Concavity, and Points of Inflection:

TODO

Note |

Prove **Periodicity**: any nonconstant rational function is not a periodic function.

$$\frac{P}{Q} = c$$

$$P - cQ = 0$$

Example |

Sketch the curve

$$y = \sqrt{x^2 + x} - x - 1$$

Chapter 02.06

Graphing with Calculus and Technology

Chapter 02.07

Optimization Problem

The methods we have learned in this chapter for finding extreme values have practical applications in many areas of life: A businessperson wants to minimize costs and maximize profits. A traveler wants to minimize transportation time. Fermat's Principle in optics states that light follows the path that takes the least time. In this section we solve such problems as maximizing areas, volumes, and profits and minimizing distances, times, and costs. In solving such practical problems the greatest challenge is often to convert the word problem into a mathematical optimization problem by setting up the function that is to be maximized or minimized. Let's recall the problem-solving principles discussed in the Principles of Problem Solving following Chapter 1 and adapt them to this situation:

Example |

A cylindrical can is to be made to hold 1 L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.

Solution 1 |

We minimize the total surface area of the cylinder.

$$A = 2\pi r^2 + 2\pi rh \quad \pi r^2 h = 1000$$

$$h = \frac{1000}{\pi r^2}$$

$$A = 2\pi r^2 + \frac{2000}{r}, \quad r > 0$$

We should find the minimum value of A .

$$A'(r) = 4\pi r - \frac{2000}{r^2}$$

Setting $A'(r) = 0$ gives

$$4\pi r = \frac{2000}{r^2} \implies r^3 = \frac{500}{\pi}$$

The only critical point is $r = \sqrt[3]{\frac{500}{\pi}}$. Checking the signs, we can prove that it is the minimum value.

Definition |

1. The **cost function** $C(x)$ is the cost of producing x units.
2. The **marginal cost** $C'(x)$ is the rate of change of cost with respect to x .
3. The **demand function** (or **price function**) $p(x)$ is the price per unit when x units are sold. (많이 팔면 더 싸짐)
4. The **revenue function** is defined by $F(x)xp(x)$
5. TODO: Fill using the ppt...

Example |

A store has been selling 200 TV monitors a week at \$ 350 each. A market survey indicates that for each \$ 10 rebate offered to buyers, the number of monitors sold will increase by 20 a week. Find the demand function and the revenue function. How large a rebate should the store offer to maximize revenue?

Solution 1 |

Let x be the number of monitors sold per week, and let $p(x)$ denote the demand function.

$$p(x) = 350 - \frac{1}{2}(x - 200) = -\frac{1}{2}x + 450, \quad x \geq 200$$

The revenue function is

$$R(x) = xp(x) = x\left(-\frac{1}{2}x + 450\right)$$

Thus $x = 450$ maximizes the revenue. Calculating the rebate:

$$\frac{1}{2}(450 - 200) = 125(\$)$$

Chapter 02.08

Optimization Problem

Sometimes the functions are too complex to calculate value from arbitrary input. We can approximate the value using the differentiation.

If f is a polynomial of degree five or higher, there is no general formula for solving $f(x) = 0$. Likewise, there is no formula for finding exact solutions of a transcendental equation such as $\cos x = x$.

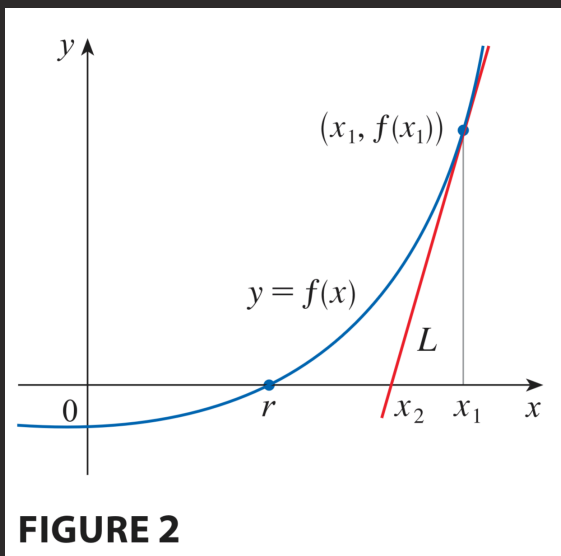
Instead of seeking exact solutions, we compute approximate solutions.

Theorem | Newton's Method

Suppose we want to solve an equation of the form $f(x) = 0$. The solutions correspond to the x-intercepts of the graph of f .

Step 1. Choose an initial approximation x_1 .

TODO: FILL BELOW



Will it converge to the solution?

Proof |

TODO

Example |

Starting with $x_1 = 0$, Starting with $x_1 = 0$

Chapter 02.09

Antiderivatives

Chapter 03

Integrals

IN CHAPTER 2 WE USED the tangent and velocity problems to introduce the derivative. In this chapter we use the area and distance problems to introduce the other central idea in calculus—the integral. The all-important relationship between the derivative and the integral is expressed in the Fundamental Theorem of Calculus, which says that differentiation and integration are in a sense inverse processes. We learn in this chapter, and in Chapters 5 and 8, how integration can be used to solve problems involving volumes, length of curves, population predictions, cardiac output, forces on a dam, work, consumer surplus, and baseball, among many others.

Chapter 03.01

The Area and Distance Problems

1 Complements

Example |

Express the limit as a definite integral:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n \frac{1}{1 + \left(\frac{i}{n}\right)^2}$$

Solution 1 |

Δx is probably $\frac{1}{n}$. x_i is probably $\frac{i}{n}$.

$$\int_0^1 \frac{1}{1 + x^2} dx$$

Example |

$$f(x) = \begin{cases} 0 & x \in \mathbb{Q} \\ 1 & x \in \mathbb{Q}^c \end{cases}$$

Show that this function is not differentiable on $[0, 1]$.

Solution 1 |

Divide $[0, 1]$ into n subintervals of equal width

$$\Delta x = \frac{1}{n} \quad x_i = \frac{i}{n}$$

Let's take sample point in the interval $[x_{i-1}, x_i]$ as a rational number. Then, the Riemann sum will be 0.

If we take sample point to be an irrational number, then Riemann sum will be 1.

Two sum do not converge to a single value. Contradiction.

Example |

Divide $[1, 2]$ into n subintervals with an equal width

$$\Delta x \quad x_i = \frac{n+i}{n}$$

Choose

$$x_i^* =$$

조대따 조대써

Chapter 03.02

The Definite Integral

Definition |

If

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x$$

We should do this using epsilon-delta, but we try to avoid this. Usually done by manipulating sample points.

Note |

Even in unequal width, $\max \Delta x_i$ must approach 0 to be an Riemann sum.

Notation |

In the notation

$$\int_a^b f(x) \, dx$$

- $\int$ is the **integral sign**.
- $f(x)$ is called the **integrand**.
- a, b is called the **limits of integral**.
- a is the **lower limit** and b is the **upper limit**.
- The procedure of calculating an integral is called **integration**.

Note |

The definite integral is a number, so it does not depend on the variable of integration. Thus we may replace x with any other letter without changing the value of integral

$$\int_a^b f(x) \, dx = \int_a^b f(r) \, dr = \int_a^b f(t) \, dt$$

A definite integral can be interpreted as a **net area**. Area under x axis is subtracted from the value.

Example |

Evaluate the Riemann sum of the function $x^3 - 6x$ on $[0, 3]$ with $n = 6$ and with the sample point chosen to be right endpoints.

Solution 1 |

With $n = 6$, we have

$$\Delta x = \frac{3 - 0}{6} = \frac{1}{2}$$

Then, the sample points are

$$x_1 = \frac{1}{2}, x_2 = 1, x_3 = \frac{3}{2}, x_4 = 2, \dots$$

Therefore,

$$A = \text{todo}$$

Theorem |

If f is continuous on $[a, b]$, or if f has only a finite number of jump discontinuities, then f is integrable on $[a, b]$.

Theorem |

If f is integrable on $[a, b]$, then

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x_i$$

where

$$\Delta x = \frac{b - a}{n} \quad \text{and} \quad x_i = a + i \Delta x$$

Example |

Evaluate $\int_0^3 (x^3 - 6x) \, dx$

Solution 1 |

Since

$$\begin{aligned}\Delta x &= \frac{3-0}{n} \quad \text{and} \quad x_i = 0 + i\Delta x \\ \int_0^3 (x^3 - 6x) \, dx &= \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left(\frac{27i^3}{n^3} - \frac{18i}{n} \right) \frac{3}{n} \\ &= \lim_{n \rightarrow \infty} \left(\frac{81}{n^4} i^3 - \dots \right) \dots\end{aligned}$$

Example |

Evaluate the following integrals by interpreting each in terms of area.

$$\int_0^1 \sqrt{1-x^2} \, dx$$

$$\int_0^1 3(x-1) \, dx$$

Solution 1 |

$$\int_0^1 \sqrt{1-x^2} \, dx$$

This is the quarter of a circle.

$$\therefore \frac{\pi \cdot 1^2}{4}$$

Theorem |

$$\int_a^b f(x) \, dx = \Delta x [f(x_1) + f(x_2) + \dots + f(x_n)]$$

where

$$\Delta x \frac{b-a}{n}$$

and x_i is midpoint of the interval.

Converges very fast, as the trapezoid method.

Theorem | Properties of the Integral

Assume f and g are continuous, and c is a constant.

- (a) $\int_b^a f(x) \, dx = - \int_a^b f(x) \, dx$
- (b) $\int_a^a f(x) \, dx = 0$
- (c) $\int_b^a c \, dx = c(b-a)$
- (d) $\int_b^a [f(x) + g(x)] \, dx = \int_b^a f(x) \, dx + \int_b^a g(x) \, dx$
- (e) $\int_b^a f(x) \, dx = \int_c^a f(x) \, dx + \int_b^c f(x) \, dx$
- (f) If $f(x) \geq 0$ for $a \leq x \leq b$, then $\int_a^b f(x) \, dx \geq 0$
- (g) If $f(x) \geq g(x)$ for $a \leq x \leq b$, then $\int_a^b f(x) \, dx \geq \int_a^b g(x) \, dx$
- (h) If $m \leq f(x) \leq M$ for $a \leq x \leq b$, then $m(b-a) \leq \int_a^b f(x) \, dx \leq M(b-a)$
- (i) $\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx$

Note |

Proof of (a) can be done by rearranging (rtl to ltr) the sample points.

(i) can be deduced from (h). (Exercise)